

Lecture 8: Sampling Distributions, Law of Large Numbers, and Central Limit Theorem

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1. Sampling Distributions, Law of Large Numbers, and Central Limit Theorem: Putting it all together

1.1 Populations and Samples

- *Population* - The entire group we want to draw conclusions about
- *Sample* - The subset of the population that we draw data from
- The sample should be a *random subset* of the population
- A *good sample* is representative of the population

1.2 Getting from sample statistic to population parameter

Two tools tie sample statistics to estimates of the true population parameters: *standard error* and *z-score*

- The standard error is a special case of the standard deviation
- The z-score is a fairly simple problem involving subtracting two numbers and dividing by the standard error **which you won't have to do** in this class

1.3 Bonuses from the Z-Score

- Bonus: the z-score is used in hypothesis testing
- Extra bonus: The rounded cutoff point for Z-Scores in hypothesis testing is really easy

$Z \geq 1.96$

$Z \geq 2$

1.4 Two important rules

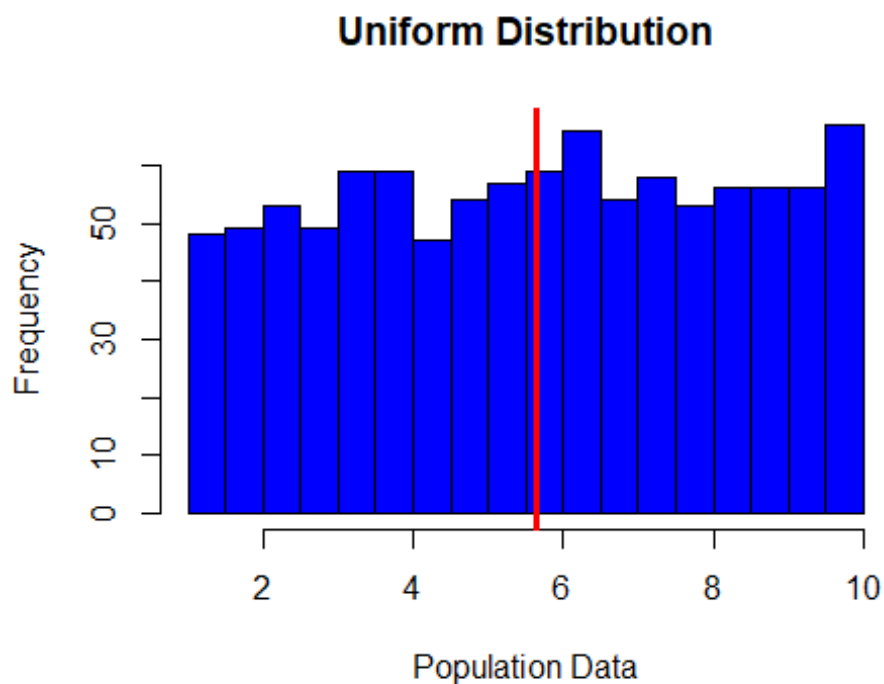
Two rules to tie the sample to probability distributions and population estimates:

- + **The Central Limit Theorem**
- + **The Law of Large Numbers**

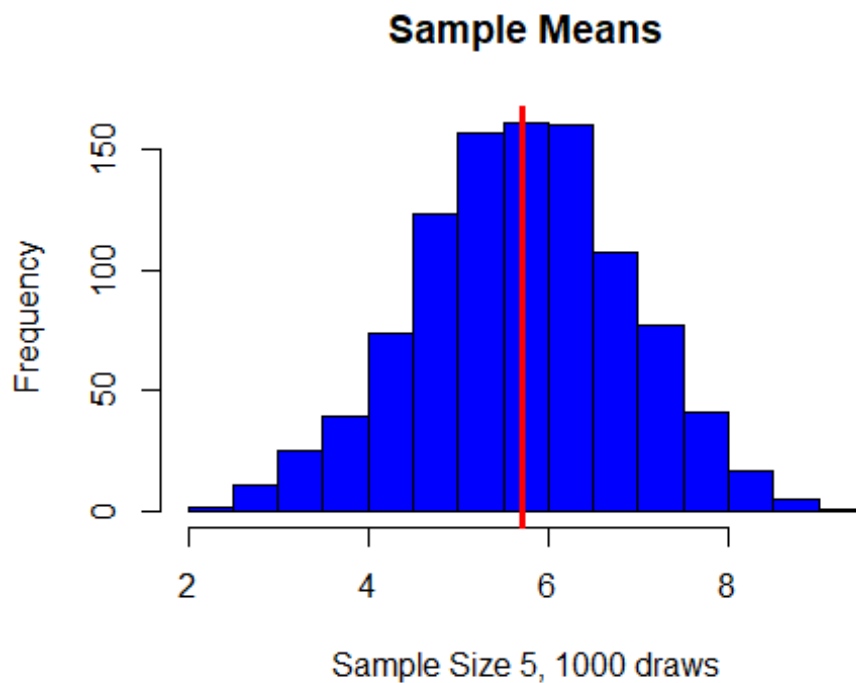
1.5 Central Limit Theorem

- **For a large number of trials, the means of the trials approach a normal distribution regardless of the underlying distribution of the data**
- This means that for a sufficient number of trials, we can apply the normal distribution to the sample means.
- This allows us to apply the 68-95-99.7 rule!

1.6 Central Limit Theorem Simulation with Uniform Distribution Data



1.7 Central Limit Theorem Simulation with Uniform Distribution Data



1.8 Central Limit Theorem Simulation with Uniform Distribution Data

```
# create a vector to store the sample means

sample_means <- c()

# create a for loop to take 1000 samples of size 30 and store the sample means

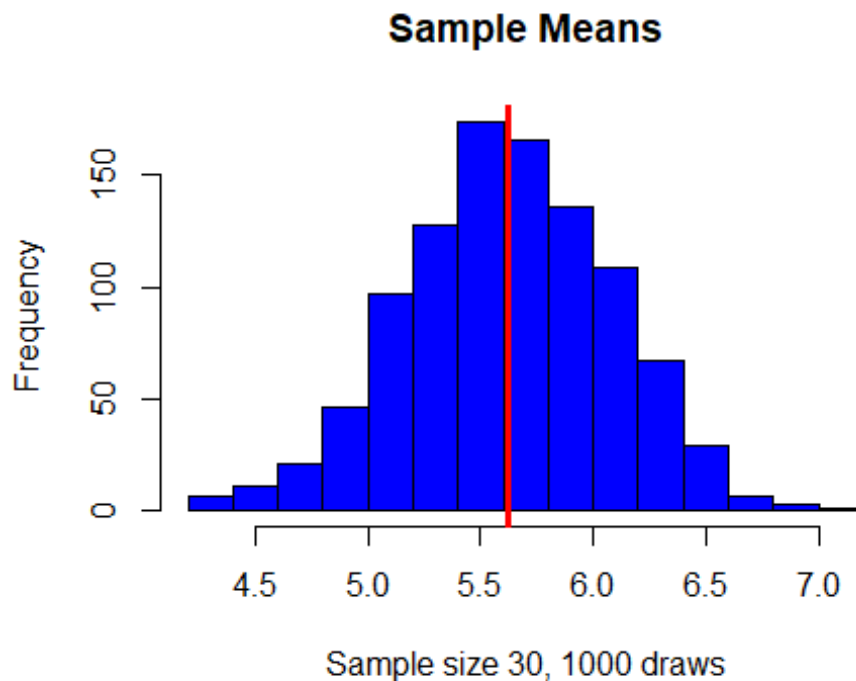
for (i in 1:1000) {
  sample_means[i] <- mean(sample(data, size=30))
}

#plot a histogram of the sample means

hist(sample_means, breaks = 20, col = "blue", main = "Sample Means", xlab =
"Sample size 30, 1000 draws")

# add a vertical line at the mean

abline(v=mean(sample_means), col="red", lwd=3)
```



1.9 Central Limit Theorem Simulation with Uniform Distribution Data

```
# create a vector to store the sample means

sample_means <- c()

# create a for loop to take 1000 samples of size 100 and store the sample means

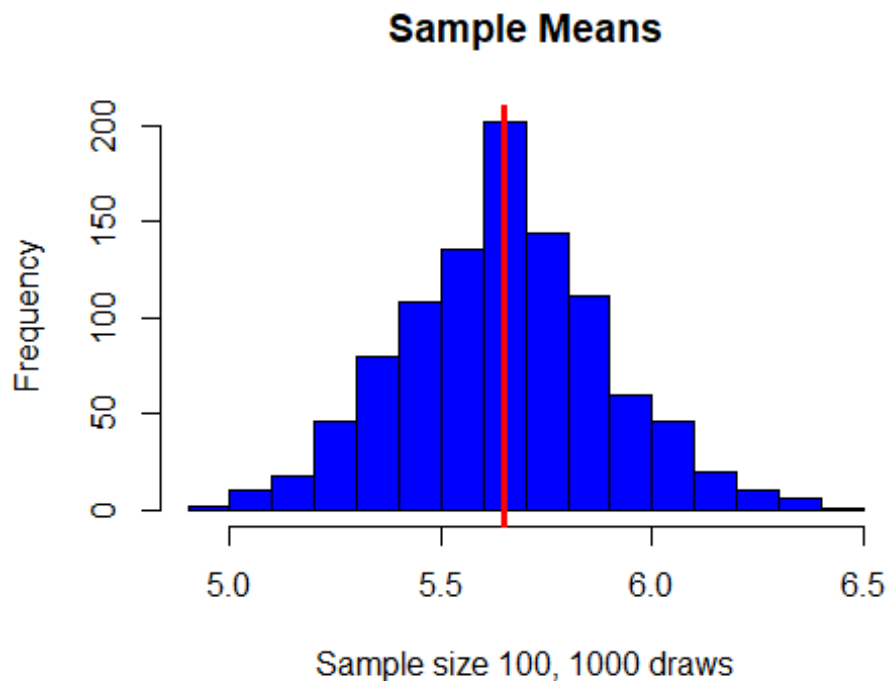
for (i in 1:1000) {
  sample_means[i] <- mean(sample(data, size=100))
}

# plot a histogram of the sample means

hist(sample_means, breaks = 20, col = "blue", main = "Sample Means", xlab =
"Sample size 100, 1000 draws")

# add a vertical line at the mean

abline(v=mean(sample_means), col="red", lwd=3)
```



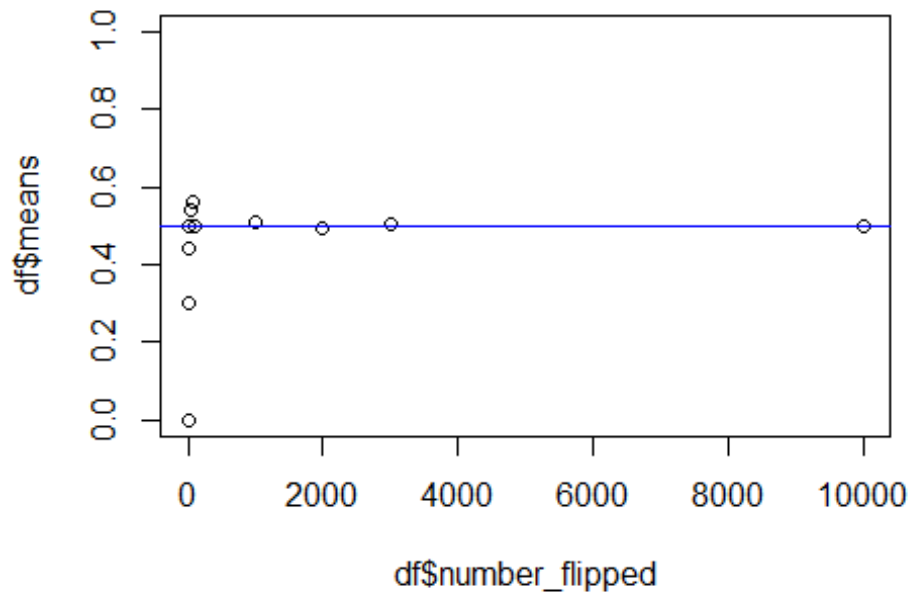
1.10 Law of Large Numbers

- **Averages from independent, identically distributed samples converge to the population means that they are estimating.**
- In its strongest form, the law states that this “almost surely” happens.
- This means that for a sufficiently large sample size, we can assume with a degree of certainty that our sample statistics accurately represent the population parameters!

1.11 Law of Large Numbers

- The mean of the sampling distribution is a good estimate of the mean of the population distribution:
- $\bar{x} = \mu$
- The standard error (the standard deviation of the sampling distribution) will be equal to the standard deviation of the population distribution divided by the sample size:
- $s = \frac{\sigma}{\sqrt{n}}$

1.12 Law of Large Numbers simulation using coin flips



The mean of a single flipped coin is 0

The mean of two flips is 0.5

The mean of ten flips is 0.3

The mean of twentyfive flips is 0.44

The mean of twenty five thousand flips is 0.5023333

1.13 What's this tell us about sample size?

- The CLT allows us to apply Z at a sample size around 30
- The sample size we need is determined by a number of things including the degree of certainty we are looking for
- Want to do some polling? This is where margin of error comes from:

$$\text{Margin of error} = z \times \frac{\sigma}{\sqrt{n}}$$

margin of error

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